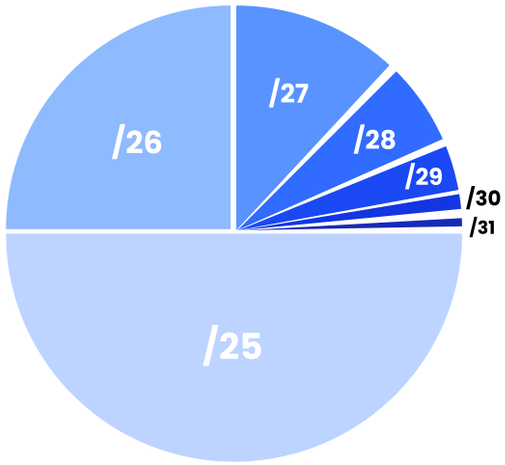
IPv4 Subnetting

Ping My Network

Subnets size



Private IPv4

Prefix	Ranges
10.0.0.0/8	10.0.0.0 – 10.255.255.255
172.16.0.0/12	172.16.0.0 – 172.31.255.255
192.168.0.0/16	192.168.0.0 – 192.168.255.255

Binary calculation

Subnet Mask		Wildcard	
255	1111 1111	**0**	0000 0000
254	1111 1110	**1**	0000 0001
252	1111 1100	**3**	0000 0011
248	1111 1000	**7**	0000 0111
240	1111 0000	**15**	0000 1111
224	1110 0000	**31**	0001 1111
192	1100 0000	**63**	0011 1111
128	1000 0000	**127**	0111 1111
0	0000 0000	**255**	1111 1111

IPv4 Classes

Classes	Ranges
A	0.0.0.0 – 127.255.255.255
B	128.0.0.0 - 191.255.255.255
C	192.0.0.0 - 223.255.255.255

Subnetting

CIDR	Subnet mask	Wildcard	Adresses
/32	255.255.255.255	0.0.0.0	1
/31	255.255.255.254	0.0.0.1	2
/30	255.255.255.252	0.0.0.3	4
/29	255.255.255.248	0.0.0.7	8
/28	255.255.255.240	0.0.0.15	16
/27	255.255.255.224	0.0.0.31	32
/26	255.255.255.192	0.0.0.63	64
/25	255.255.255.128	0.0.0.127	128
/24	255.255.255.0	0.0.0.255	256
/23	255.255.254.0	0.0.1.255	512
/22	255.255.252.0	0.0.3.255	1,024
/21	255.255.248.0	0.0.7.255	2,048
/20	255.255.240.0	0.0.15.255	4,096
/19	255.255.224.0	0.0.31.255	8,192
/18	255.255.192.0	0.0.63.255	16,384
/17	255.255.128.0	0.0.127.255	32,768
/16	255.255.0.0	0.0.255.255	65,536
/15	255.254.0.0	0.1.255.255	131,072
/14	255.252.0.0	0.3.255.255	262,144
/13	255.248.0.0	0.7.255.255	524,288
/12	255.240.0.0	0.15.255.255	1,048,576
/11	255.224.0.0	0.31.255.255	2,097,152
/10	255.192.0.0	0.63.255.255	4,194,304
/9	255.128.0.0	0.127.255.255	8,388,608
/8	255.0.0.0	0.255.255.255	16,777,216
/7	254.0.0.0	1.255.255.255	33,554,432
/6	252.0.0.0	3.255.255.255	67,108,864
/5	248.0.0.0	7.255.255.255	134,217,728
/4	240.0.0.0	15.255.255.255	268,435,456
/3	224.0.0.0	31.255.255.255	536,870,912
/2	192.0.0.0	63.255.255.255	1,073,741,824
/1	128.0.0.0	127.255.255.255	2,147,483,648
/0	0.0.0.0	255.255.255.255	4,294,967,296